

CLOUD FUNDAMENTALS – AZURE

1. Objective

The objective of this assignment was to gain practical experience with Microsoft Azure by deploying a web application using **Azure App Service**, registering an application with **Microsoft Entra ID**, and configuring access control using **Azure Role-Based Access Control (RBAC)**.

The main activities performed were:

- Deploying a sample web application using Azure App Service.
- Registering the application in Microsoft Entra ID.
- Configuring application identity and authentication-related settings.
- Reviewing Azure RBAC access for an Azure Storage Account.
- Verifying the deployed application and Azure resources.

2. Azure App Service Deployment

Azure App Service is a fully managed platform for hosting web applications and APIs without requiring users to manage the underlying infrastructure.

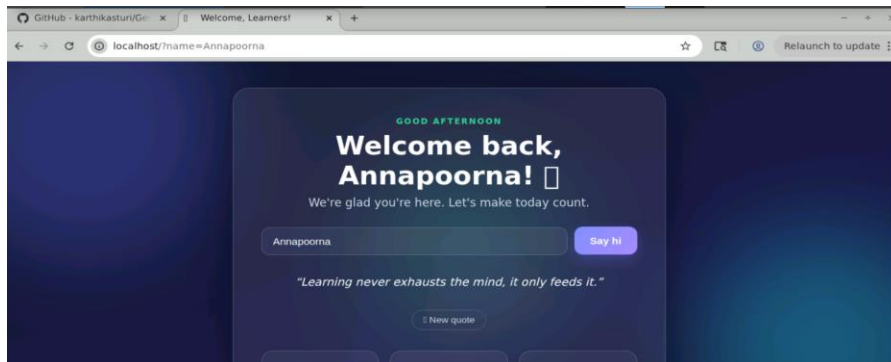
A sample web application was developed and tested locally before deployment. The application accepts a name as an input and displays a personalized welcome message.

The application was verified locally using the following format:

localhost/?name=Annapoorna

The application displayed a personalized welcome page with the message “**Welcome back, Annapoorna!**”, confirming that the application was functioning correctly.

Figure 1: Sample Web Application



Deployment Procedure

1. Create and test the sample web application locally.
2. Verify that the application is running correctly.
3. Create an **Azure App Service** resource in the Azure Portal.
4. Configure the application name, runtime, region, and required hosting settings.
5. Configure the required deployment source.
6. Deploy the application to Azure App Service.
7. Access the App Service URL through a web browser.
8. Verify that the deployed application produces the expected output.

Azure App Service provides a managed hosting environment, making it possible to deploy and run web applications without directly managing virtual machines or operating-system infrastructure.

3. Microsoft Entra ID – Application Registration

Microsoft Entra ID provides identity and access management capabilities for applications and users. Application registration creates an identity for an application and allows authentication and authorization to be configured.

An application registration named:

adm-itc-annapoorna-entraID

was created in the Azure environment.

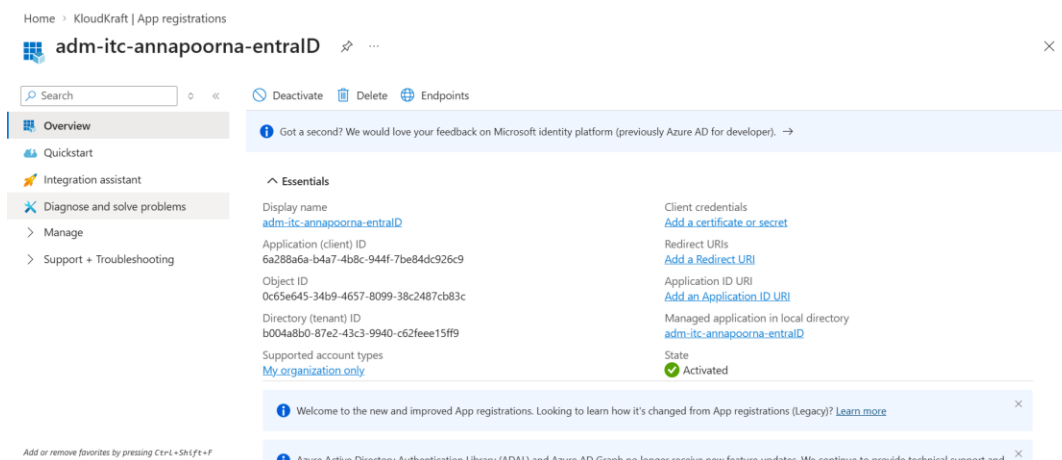
The application registration was successfully activated and its configuration was reviewed through the Azure Portal.

Application Registration Details

Parameter	Value
Display Name	adm-its-annapoorna-entraID
Application Type	App Registration
Supported Account Type	My organization only
State	Activated
Client Credentials	Configurable
Redirect URI	Configurable
Application ID URI	Configurable

The Azure Portal displays the application registration along with its client ID, object ID, directory/tenant ID, and application configuration options.

Figure 2: Microsoft Entra ID Application Registration



The application registration provides an identity for the application within the organization's Microsoft Entra ID environment. This identity can subsequently be used to configure authentication and authorization for the application.

Registration Procedure

1. Open the **Azure Portal**.
2. Navigate to **Microsoft Entra ID**.
3. Select **App registrations**.
4. Select **New registration**.
5. Enter the application name **adm-its-annapoorna-entraID**.
6. Select the appropriate supported account type.
7. Configure the redirect URI if required by the application.

8. Register the application.
9. Open the newly created application registration.
10. Verify the Application (Client) ID, Directory (Tenant) ID, and application status.

The registration was configured for **accounts in the organization only**, providing a controlled authentication scope.

4. Azure RBAC – Access Control

Azure Role-Based Access Control (RBAC) is used to manage access to Azure resources by assigning appropriate roles to users, groups, service principals, and managed identities.

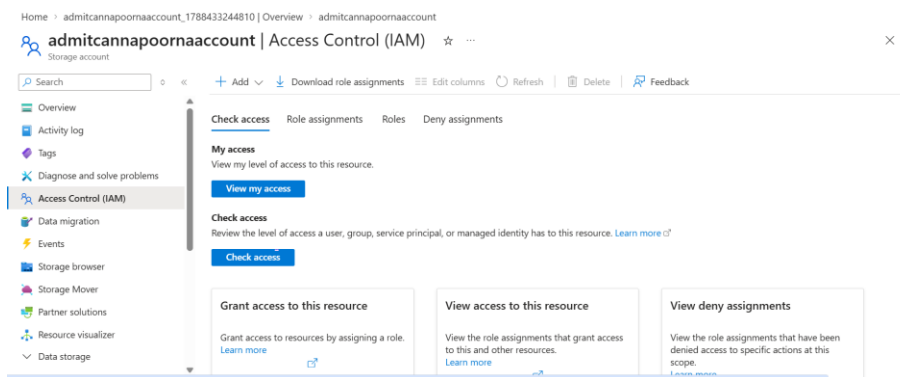
As part of the assignment, access control for an Azure Storage Account was reviewed.

The storage account shown in the configuration is:

admitcannapoornaaccount

The **Access Control (IAM)** section provides options to check existing access, view role assignments, grant access, and review deny assignments.

Figure 3: Azure Storage Account – Access Control (IAM)



The screenshot shows the Access Control (IAM) section of the storage account. It provides functionality to review access levels and manage role assignments for the resource.

RBAC Procedure

1. Open the Azure Portal.
2. Navigate to the required Storage Account.
3. Open **Access Control (IAM)**.

4. Review the **Role assignments** section.
5. Use **Check access** to review the permissions of a user, group, service principal, or managed identity.
6. Assign an appropriate Azure RBAC role when access is required.
7. Verify that the assigned role provides only the permissions necessary for the intended operation.

RBAC follows the principle of assigning permissions according to the user's or application's responsibilities. This helps avoid granting unnecessary access to Azure resources.

5. Verification and Result

The completed activities were verified through both the local application and Azure Portal.

The following results were obtained:

- The sample web application was successfully executed locally.
- The application accepted the name parameter and displayed a personalized welcome message for Annapoorna.
- The application registration `adm-its-annapoorna-entraID` was successfully created and shown as **Activated** in Microsoft Entra ID.
- The application registration was configured for organizational accounts.
- The Azure Storage Account `admitcannapoornaaccount` was accessed through its **Access Control (IAM)** section.
- Azure RBAC provides mechanisms to check access and manage role assignments for the storage resource.

6. Conclusion

The assignment provided hands-on experience with **Azure App Service, Microsoft Entra ID, and Azure Role-Based Access Control**.

A sample web application was developed and verified locally before deployment. An application registration named `adm-its-annapoorna-entraID` was created in Microsoft Entra ID to provide an identity for the application. Azure Access Control (IAM) was also reviewed for the storage account to understand how role assignments and resource-level permissions are managed.

Overall, the assignment provided practical understanding of **Azure application hosting, application registration, authentication infrastructure, and role-based access control**, which are important components of developing and deploying secure cloud applications.

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